

Non-Anticipative Hierarchical Event Diagnosis and Physics-Guided Candidate Localization from Sparse PMU Measurements

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Abstract—Sparse phasor measurement unit (PMU) coverage makes event localization an inverse source-inference problem because the initiating BUS or LINE is often unmeasured and telemetry loss can resemble a physical transient. We evaluate three estimator structures under the same non-anticipative protocol: flat output heads, typed BUS/LINE/PMU routing, and candidate-conditioned localization with an impedance-derived network descriptor. The study uses 690 five-second ANDES simulations of the IEEE 39-bus system observed by eight PMUs. Complete trajectories are separated across training, validation, and test partitions, and each timestamp-level decision uses only measurements available at or before that timestamp through a trailing 3-s representation. Flat obtains 0.977 detection F1 and 0.535 row-level physical Top-1. Typed routing obtains 0.965 and 0.496, respectively, while reducing serialized size from 98.3 to 28.1 MB and model-only scoring time from 0.051 to 0.025 s per input minute. Its offline scenario-level physical Top-1 is 0.614 versus 0.590 for flat, with a paired interval that includes zero. The candidate-conditioned design gives no aggregate localization gain and reduces line-outage Top-1 in the tested configuration. Missing-only detection and load localization remain the principal failure cases. Under the evaluated split, typed routing is a compact structured alternative rather than an accuracy improvement, and a single static impedance descriptor is not sufficient as a universal physical source representation. The reported estimates apply to new trajectories at known assets.

Index Terms—ANDES, event detection, event localization, ExtraTrees, phasor measurement units, physics-guided machine learning, synchrophasors.

I. INTRODUCTION

PHASOR measurement units provide time-synchronized voltage and current phasors together with frequency-related measurements, enabling wide-area observation of transmission-system dynamics [1], [2]. Localization is harder than event detection when only a small fraction of buses is instrumented: a disturbance at an unobserved bus or line must be inferred from the response measured elsewhere in the network. In the SGSMA 2026 problem, eight PMUs observe the IEEE 39-bus

system while the required output includes event type and origin [3].

Sparse sensing creates two distinct ambiguities. Different physical sources can produce similar responses at the available PMUs, so state observability does not imply source identifiability [4], [5]. Separately, missing or corrupted telemetry can resemble the onset of a physical event unless the measurement process is modeled explicitly [6]–[8]. BUS, LINE, and PMU origins therefore should not be treated as interchangeable labels.

The competition prototype used complete 30-s windows, current and voltage phasors, ROCOF, robust statistics, multi-scale rolling features, adaptive residuals, and graph-temporal summaries. That design was appropriate for retrospective competition scoring, but a complete event window is not an appropriate information set for timestamp-level online inference. The formulation evaluated here is intentionally different: it uses a trailing 3-s history, 136 past-only variables, whole-trajectory train/validation/test separation, and one prediction per timestamp. Section III makes this transition explicit so that results from the competition prototype are not used to validate the non-anticipative formulation.

The matched experiment separates three issues that are often conflated in sparse-PMU studies: the effect of typed output structure, the effect of candidate-conditioned network information, and the event families that remain poorly identifiable from the prescribed eight-PMU view. The contribution is methodological as much as algorithmic: every timestamp-level prediction obeys the same non-anticipation rule, the three estimator structures share the same trajectory partitions and total tree budget, and failure modes are reported by event family rather than hidden inside one aggregate score. The candidate-conditioned model is evaluated as a complete architecture; the present experiment does not isolate the impedance descriptor by itself.

II. RELATED WORK AND POSITIONING

State estimation, missing-measurement reconstruction, and event localization use similar synchrophasor streams but estimate different latent quantities. Incomplete observability, communication latency, and synchronization errors therefore affect them differently [9]. We focus on transmission-system event diagnosis and use state-estimation work where it clarifies the information that sparse synchronized measurements can retain.

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The initial prototype underlying this study was developed for the Synchrophasor Anomaly Detection Competition at the 5th International Conference on Smart Grid Synchronized Measurements and Analytics (SGSMA 2026), Santiago, Chile, June 1–4, 2026.

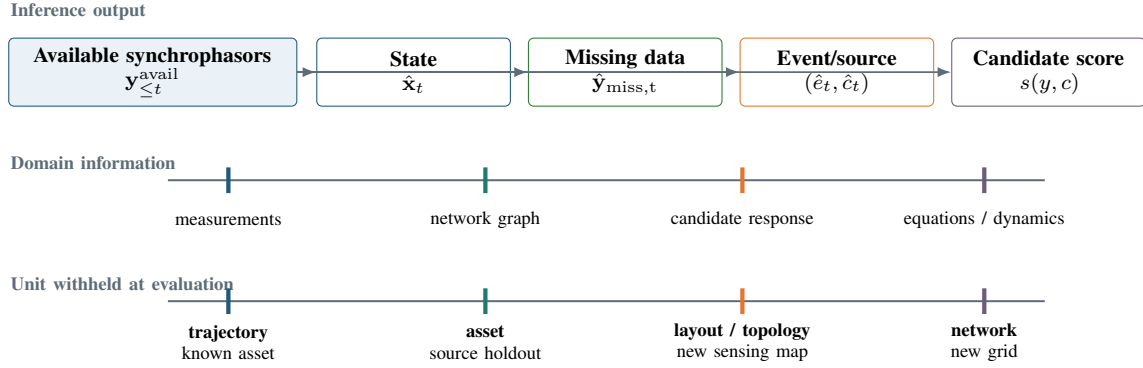


Fig. 1. Three distinctions needed to compare sparse-PMU studies: the latent quantity being inferred, the physical/network information supplied to the estimator, and the independent unit withheld during evaluation. These axes are separate; a graph-based state estimator, for example, need not test source-asset transfer.

A. Event Identification from PMU Streams

Integrated PMU diagnosis is well established. Pandey *et al.* combine detection, classification, localization, topology information, and bad-data processing [10]. Aalam and Shubhanga use wavelet energy and statistical indicators on multiple PMUs, including ISO-NE field records [11]. Long field records introduce drift and operating-condition variation: Yuan *et al.* use two years of PMU measurements and event logs [12], Ahmed *et al.* study drifting synchrophasor streams [13], and Cheng *et al.* develop an online GAN detector [14]. Limited labels have also been addressed through structural tensor learning [15].

Yuan, Wang, and Wang infer PMU-to-PMU interactions from data rather than fixing the transmission graph [16]. A learned interaction graph may be predictive without representing admittance, transfer impedance, or contingency sensitivity. The review in [17] covers network-aware features, model-based constraints, and equation-coupled learning. DynamoPMU, for example, extracts nonlinear dynamical structure from μ PMU trajectories [18]. A learned interaction graph and an equation-constrained estimator encode different assumptions about the network.

B. Sparse Localization, Placement, and Candidate Models

Sensor placement is part of the localization problem. Zhao, Goldsmith, and Poor select PMU locations so that projected line-outage signatures remain separated [19]. Li, Deka, Chertkov, and Wang combine CNN-based faulted-line localization with PMU placement on the IEEE 39- and 68-bus systems [20]. Jamei *et al.* analyze the statistical distinguishability of competing fault hypotheses under sparse sensing [5]. Yildiz and Abur study fault location with few PMUs and later optimize sparse PMU placement for event identification [21], [22].

Azizi *et al.* compare candidate generation-loss locations against network-predicted PMU responses [23] and use candidate-dependent network relationships again for line-outage monitoring [24]. Sánchez Cortés *et al.* use a related physical-consistency argument for generation-loss estimation under high renewable penetration [25]. The matched quantity is event dependent: an opened line changes network redistribution, whereas a generation loss appears through an injection imbalance and subsequent electromechanical response.

Matching Networks, Prototypical Networks, and Relation Networks likewise compare an observation with support examples or class representations [26]–[28]. Here the candidate is a physical BUS, LINE, or PMU asset, so the representation must carry enough network information to separate competing sources.

C. State and Measurement Reconstruction under Sparse Observation

State-estimation studies provide useful robustness references, although their latent target is an electrical state rather than an event origin. Classical DSE combines dynamic models, recursive estimation, and observability conditions [30]; Qi, Sun, and Kang connect PMU placement to DSE observability through an empirical Gramian [31]. Recent work augments this pipeline with graph and factor-graph message passing [32], [33], explicit network relations [34], and topology-aware GNNs for PMU-unobservable or degraded systems [35]. Missing PMU values can be reconstructed through tensor decomposition [36] or deep learning with magnitude-trend decoupling [37]. These methods recover electrical states or measurements rather than an event origin.

D. Generalization, Integrity, and Evaluation Scope

Trajectory, asset, sensing-layout, and cross-network holdouts test different forms of generalization. Li, Ma, and Weng address cross-grid transfer by aligning event zones across different measurement spaces [29]; Wu *et al.* study zero-shot transfer across topology reconfigurations [38]. A new trajectory at a known asset, a source withheld from fitting, a new PMU layout, and a different network are four different evaluation units. The experiment reported here addresses only the first.

Telemetry failures change the observation process rather than the physical grid state. Liu *et al.* show that imperfect real-world PMU measurements materially affect event classification [7]; event-signature and PNNL data resources document both physical disturbances and data-quality problems [39]–[41]. A failed PMU changes the observation operator and may itself be the location to be inferred.

TABLE I

REPRESENTATIVE EVENT-DIAGNOSIS LITERATURE AND THE PHYSICAL MEANING OF THE REPORTED OUTPUT. ACCURACY VALUES ARE NOT COMPARED BECAUSE EVENT INVENTORIES, CANDIDATE SPACES, AND SENSING LAYOUTS DIFFER.

Work	Output	System / sensing	Network or physical information	Evaluation unit
Zhao–Goldsmith–Poor [19]	line-outage signature	IEEE 14/24/30; PMU placement	projected outage-signature separation	placement / separability
Li <i>et al.</i> [20]	faulted line	IEEE 39/68; optimized PMUs	learned voltage signatures + placement	operating / sensing variations
Jamei <i>et al.</i> [5]	fault hypothesis	sparse-PMU fault study	information-theoretic distinguishability	placement / observability
Azizi <i>et al.</i> [23], [24]	generator or line candidate	transmission systems; sparse PMUs	candidate-specific network response	candidates in studied systems
Yuan <i>et al.</i> [12], [16]	event type	large real PMU corpora	temporal model; learned PMU interactions	temporal field data
Li–Ma–Weng [29]	event type + event zone	source and target grids	latent-domain / label-space alignment	cross-grid transfer
Yildiz–Abur [21], [22]	fault / event location	transmission test systems; few PMUs	CNN signatures + PMU placement	few-PMU / placement studies
This study	BUS / LINE / PMU origin + event type	IEEE 39; 8/39 PMUs	typed routing + candidate network descriptor	trajectory holdout; known assets

III. STUDY DESIGN, POWER-SYSTEM MODEL, AND DATA

A. Prototype-to-Evaluation Transition

The competition implementation and the experiment reported in this paper are related but not identical statistical objects. Table II records the transition. The earlier implementation was designed to score complete competition windows and used a large retrospective feature bank. The evaluated formulation was rebuilt around a timestamp-level information contract so that future samples cannot enter an earlier prediction.

B. IEEE 39-Bus Network and PMU Placement

The study uses the 39-bus New England system, a 100-MVA, 60-Hz transmission test network with 39 buses and 10 generator buses [42], [43]. The SGSMA measurement layout places PMUs at

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \quad (1)$$

The eight streams are time aligned and provide three-phase voltage and current phasors, frequency, ROCOF, availability, and timestamp information [3]. The exact PMU-to-bus mapping is PMU1–Bus39, PMU2–Bus29, PMU3–Bus10, PMU4–Bus22, PMU5–Bus19, PMU6–Bus2, PMU7–Bus5, and PMU8–Bus6.

Figure 2 shows the electrical connectivity used by the simulation and the network descriptor. The drawing is schematic rather than geographic; solid edges denote transmission branches and dashed edges the generator step-up transformer connections represented in the standard case data.

C. Network Equations and Physical Event Signatures

At the network level, Kirchhoff’s current law gives

$$\mathbf{i} = Y_{\text{bus}} \mathbf{v}, \quad (2)$$

and the complex power injection at bus i is

$$S_i = P_i + jQ_i = V_i \left(\sum_{j=1}^N Y_{ij} V_j \right)^*. \quad (3)$$

Writing $Y_{ij} = G_{ij} + jB_{ij}$ and $\theta_{ij} = \theta_i - \theta_j$ gives the familiar nonlinear AC balance

$$P_i = |V_i| \sum_j |V_j| (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}), \quad (4)$$

$$Q_i = |V_i| \sum_j |V_j| (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}). \quad (5)$$

These equations explain why a disturbance at an unobserved bus can still be inferred from correlated changes at the PMU buses.

For small current-injection perturbations around an operating point,

$$\Delta \mathbf{v} \approx Z_{\text{bus}} \Delta \mathbf{i}, \quad Z_{\text{bus}} = Y_{\text{bus}}^+, \quad (6)$$

which motivates the impedance-derived candidate descriptor used later. Electromechanical frequency changes follow active-power imbalance. A classical first-swing interpretation is

$$\frac{2H_i}{\omega_s} \dot{\Delta \omega}_i = \Delta P_{m,i} - \Delta P_{e,i} - D_i \Delta \omega_i, \quad \dot{\delta}_i = \Delta \omega_i, \quad (7)$$

so frequency and ROCOF carry information about generation/load imbalance [44]. Line outages are different: they redistribute branch flows and phase-angle differences. For interpretation, the post-outage flow change on monitored line ℓ due to outage m can be written in the standard sensitivity form

$$P_{\ell}^{\text{post}} \approx P_{\ell}^{\text{pre}} + \text{LODF}_{\ell,m} P_m^{\text{pre}}. \quad (8)$$

The implemented localizer does not use LODF features. Equation (8) shows why a single static impedance descriptor need not be equally informative for faults, line outages, and injection changes.

D. ANDES Scenario Construction

The dynamic trajectories are generated with ANDES on the IEEE 39-bus system [45]. ANDES represents the model as a differential-algebraic system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{u}; \boldsymbol{\eta}), \quad (9)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{u}; \boldsymbol{\eta}), \quad (10)$$

where $\mathbf{x}$ collects dynamic device states, $\mathbf{y}$ algebraic network variables, $\mathbf{u}$ disturbances, and $\boldsymbol{\eta}$ model parameters. Physical scenarios modify network or injection variables and propagate through the DAE solution. Missing and bad-data scenarios are applied to the sampled measurement stream so that telemetry failures remain distinct from physical grid disturbances.

All numerical results in Sec. VII use a separate 690-trajectory simulation set rather than the original 90-min competition stream. The simulation set contains 138 target keys with five replicas each. Every event begins at 2 s; faults last 0.2 s, persistent generation/load changes continue through the record, and missing/corrupted intervals last 0.7 s. Event severity,

TABLE II
COMPETITION PROTOTYPE AND NON-ANTICIPATIVE FORMULATION EVALUATED IN THIS PAPER. THE DISTINCTION PREVENTS RETROSPECTIVE COMPETITION RESULTS FROM BEING USED AS EVIDENCE FOR THE TIMESTAMP-LEVEL EXPERIMENT.

Element	Competition prototype	Evaluated formulation
Decision unit	complete 30-s event window; one segment decision written at timestamp resolution	one decision at each timestamp from a trailing 3-s history
Temporal information	retrospective pre/early/mid/late/full segments	samples at or before decision time only
Signals	voltage/current magnitudes and angles, frequency, ROCOF, availability	voltage magnitude, unwrapped voltage angle, frequency, availability
Feature families	robust deviation, derivatives, multiscale rolling summaries, RLS/adaptive residuals, phase/power proxies, graph-temporal summaries	current value, initial-reference departure, trailing mean/std./slope, missing fraction, fixed scalar summary
Feature dimension	45,162 archived variables	136 variables
Network use	graph-temporal pair features and electrical-distance weights	candidate-to-PMU impedance descriptor from Z_{bus}
Development data	synthetic competition windows with RAW0001 transfer analysis	690 ANDES trajectories; 414/138/138 grouped train/validation/test split
Scientific purpose	competition scoring pipeline	non-anticipative comparison of flat, typed, and candidate-conditioned estimators

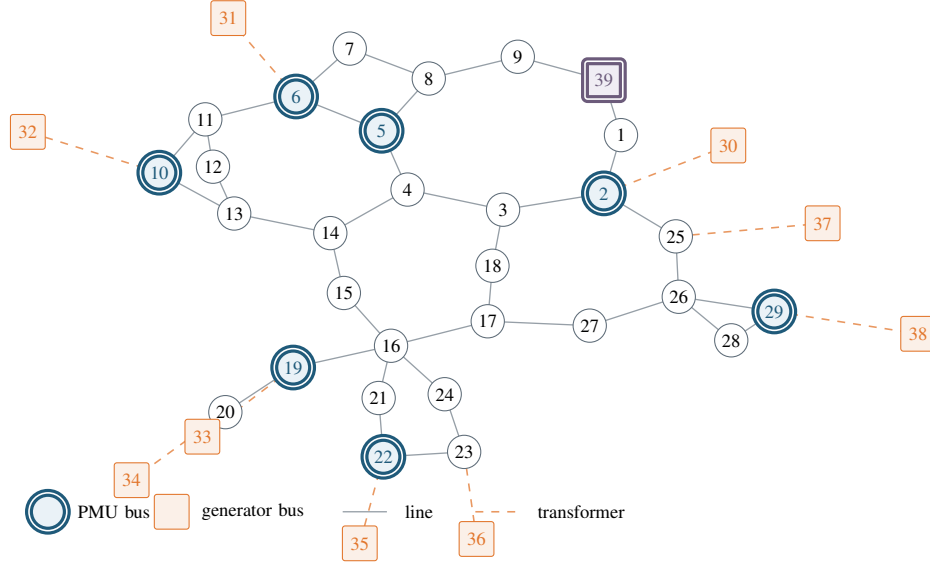


Fig. 2. Schematic one-line connectivity of the IEEE 39-bus system used in the study. Double-outline nodes mark the eight PMU buses; rectangular nodes are generator buses. Connectivity follows the standard case39 branch data; the layout is not geographic.

TABLE III
PHYSICAL INTERPRETATION OF EVENT FAMILIES AND NETWORK RELATIONS RELEVANT TO LOCALIZATION.

Family	Dominant physical effect	Observable PMU evidence	Network relation
Fault	short-circuit current and voltage depression	fast $ V $ /angle departure; current is available in the source stream	transfer-impedance / fault-response relation
Line outage	redistribution of branch flows	voltage-angle shift and multi-PMU redistribution; current phasors are available	PTDF/LODF or post-contingency response
Generation Load	active-power imbalance active/reactive injection change	frequency, ROCOF, angle and voltage response voltage/angle/frequency response	injection-to-state sensitivity + machine dynamics AC/Jacobian injection sensitivity
Missing data	observation loss, not a grid state change	DATA_PRESENT=0 and NaNs at one PMU	time-varying observation operator
Bad data	inconsistent telemetry while stream remains present	isolated channel/PMU inconsistency	measurement-integrity model

operating scale, and simulator seed vary across replicas. Scale, noise, and sampling rate are calibrated from normal rows in RAW0001; RAW labels are not used for fitting the reported models or selecting their test thresholds.

The competition specification defines Event 6 generically as missing data concurrent with a physical event. In the simulation set used here, Event 6 is instantiated as generation change plus missing data, so its physical source is a BUS. The competition specification describes Event 8 as unknown/ambiguous; the simulation set instantiates it as a closed mixed physical class and does not evaluate open-set recognition. These distinctions

keep the reported estimand tied to the actual simulated labels.

The simulated outputs are sampled trajectories, not a complete standard-compliant PMU/PDC chain. Time-synchronization error, packet reordering, communication buffering, estimator response, and other instrument-level effects are not reproduced.

TABLE IV
SIMULATION-SET INVENTORY. EACH TARGET KEY HAS FIVE REPLICAS.

e	Family	Keys	Scenarios
0	normal	1	5
1	bus fault	39	195
2	line outage	34	170
3	generation change	10	50
4	load change	18	90
5	missing data	8	40
6	generation + missing	10	50
7	bad data	8	40
8	mixed physical class	10	50
Total		138	690

TABLE V
MEASUREMENT CONTRACT AND CAUSAL INPUT SET.

Measurement family	Available	Used
Voltage magnitude / angle	yes	yes
Current magnitude / angle	yes	no
Frequency	yes	yes
ROCOF	yes	no
Availability / missingness	yes	yes
Timestamp	yes	alignment/windowing

TABLE VI
EVENT-COMPATIBLE ORIGIN SPACES USED IN THE EXPERIMENT.

e	Event	Physical candidates	Integrity
0	normal	—	—
1	fault	39 BUS	—
2	line outage	34 LINE	—
3	generation change	10 BUS	—
4	load change	18 BUS	—
5	missing data	—	8 PMU
6	generation + missing	10 BUS	8 PMU
7	bad data	—	8 PMU
8	mixed physical class	10 BUS	optional

IV. PROBLEM FORMULATION

A. Structured Output and Candidate Spaces

At timestamp t , the required output is

$$\hat{\mathbf{z}}_t = (\hat{e}_t, \hat{e}_t^{\text{phys}}, \hat{e}_t^{\text{int}}), \quad (11)$$

where either origin is empty when it does not apply. The simulation set contains 39 fault BUS candidates, 34 LINE candidates, 10 generation BUS candidates, 18 load BUS candidates, and eight PMU integrity sites. The 34 line-outage candidates are the physical transmission branches in the case39 branch table with unit tap and zero phase shift; transformer branches are excluded from the line-outage label space.

Event 6 is instantiated in this simulation set as generation change plus missing data, so its physical origin is a BUS and its integrity origin is a PMU. Event 8 is evaluated as a closed mixed physical class rather than an open-set class.

B. Non-Anticipative Observation Contract

Let

$$\mathcal{Y}_t = \{\mathbf{y}_{\mathcal{P}_\tau, \tau} : \tau \leq t\} \quad (12)$$

denote the measurements available up to decision time t , with $\mathcal{P}_\tau \subseteq \mathcal{P}$ under dropout. The estimator is non-anticipative when

$$\hat{\mathbf{z}}_t = f(\Phi(\mathcal{Y}_t)) \quad (13)$$

is unchanged by any modification to samples after t .

If ρ_t is the abnormal forest score, the first-stage decision is

$$\hat{d}_t = \mathbf{1}\{\rho_t \geq \gamma\}, \quad (14)$$

where γ is selected on validation data and frozen before test evaluation. Once an event family is selected, localization is restricted to its compatible candidate set. A fixed class head predicts within that set, while the candidate-conditioned variant returns

$$\hat{c}_t = \arg \max_{c \in \mathcal{C}_{\hat{e}_t}} s_\theta(c \mid \mathbf{x}_t). \quad (15)$$

Every target asset appears in training, validation, and test. The independent test unit is therefore a new trajectory at a known source. This split does not estimate performance for an unseen BUS or LINE, a new PMU layout, or another network.

V. HIERARCHICAL EVENT DIAGNOSIS AND CANDIDATE LOCALIZATION

A. Causal Feature Construction and Missing-Data Preprocessing

At timestamp t , the extractor reads the trailing interval

$$\mathcal{W}_t = [\max(t - 3 \text{ s}, t_0), t]. \quad (16)$$

For retained signal $s \in \{|V|, \theta, f\}$ at PMU p , the first finite value defines a causal scenario reference

$$b_{p,s} = x_{p,s} \left(t_{p,s}^{(0)} \right). \quad (17)$$

Each signal contributes

$$\phi_{p,s}(t) = [x_{p,s}(t), x_{p,s}(t) - b_{p,s}, \mu_{p,s}(t), \sigma_{p,s}(t), \beta_{p,s}(t)], \quad (18)$$

where μ , σ , and β are the trailing mean, standard deviation, and least-squares slope over $\mathcal{W}_t$. A trailing missing fraction $m_p(t)$ and a deterministic scalar disturbance summary $q_p(t)$ complete the PMU block,

$$\mathbf{x}_{p,t} = [\phi_{p,|V|}(t), \phi_{p,\theta}(t), \phi_{p,f}(t), m_p(t), q_p(t)] \in \mathbb{R}^{17}, \quad (19)$$

and

$$\mathbf{x}_t = [\mathbf{x}_{p,t}]_{p \in \mathcal{P}} \in \mathbb{R}^{136}. \quad (20)$$

The scalar summary is fixed across all compared models.

Missingness is preserved before numerical filling. The availability/NaN indicators are formed first; remaining non-finite feature values presented to the tree models are median-imputed using statistics estimated on the training partition. This follows the tabular preprocessing logic of the original tree pipeline while preserving missingness as predictive evidence. A future-change unit test modifies samples strictly after t and verifies that all features at $\tau \leq t$ remain unchanged.

The five statistics play complementary roles. Reference departure and trailing mean represent persistent level changes, trailing spread responds to transient variability, and the local slope represents ramps or settling dynamics. The missing fraction is computed before imputation so that the model retains explicit evidence about the observation process. All imputation statistics are estimated from the training partition only.

The competition prototype used robust deviation, sustained-versus-isolated anomaly shape, multiscale rolling summaries, adaptive residuals, and graph-temporal propagation cues. The

TABLE VII
CAUSAL FEATURE BLOCK PER PMU.

Signal	Statistics	Count
$ V $	current, departure, mean, std., slope	5
θ	current, departure, mean, std., slope	5
f	current, departure, mean, std., slope	5
Availability	trailing missing fraction	1
Disturbance summary	deterministic scalar	1
Total per PMU		17
Total, 8 PMUs		136

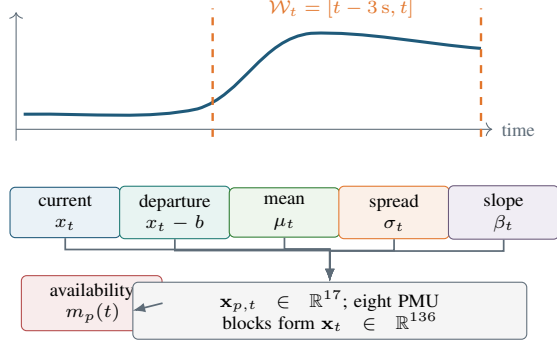


Fig. 3. Causal feature construction for one retained PMU signal. The trailing 3-s window produces current value, initial-reference departure, local mean, spread, and slope; availability is computed before imputation. The same statistics are evaluated for $|V|$, θ , and f at each of the eight PMUs.

non-anticipative formulation retains the underlying ideas of level change, transient spread, slope, missingness, and network-relative evidence, but reduces the representation so that each reported output can be computed without future samples.

B. Ensemble Event and Origin Heads

The learning layer uses Extremely Randomized Trees [46], [47]. At each node, ExtraTrees evaluates random feature/threshold proposals and selects a split according to impurity reduction. If π_k denotes the class fraction in a node, the Gini impurity is

$$G = 1 - \sum_k \pi_k^2. \quad (21)$$

For tree m , let $p_m(k | \mathbf{x})$ be the class fraction in the reached leaf. The ensemble vote is

$$p_{\text{ET}}(k | \mathbf{x}) = \frac{1}{M} \sum_{m=1}^M p_m(k | \mathbf{x}). \quad (22)$$

The vote fractions are used for ranking and thresholding only.

The flat architecture fits event, physical-location, and integrity-location heads. The typed architecture separates abnormal detection from event identification and then uses event-specific location heads. The candidate-ranking architecture retains the typed event stage and replaces the fixed location heads with shared candidate scorers. The same trajectory partitions and total 360-tree budget are used for all three variants.

The typed design preserves the prototype's separation between measurement-integrity failures and physical origins. Missing or corrupted telemetry is assigned to a PMU origin space, whereas physical events are routed to BUS or LINE

candidates. ExtraTrees is used to learn interactions among the engineered statistics without requiring a manually tuned rule for every event family.

C. Network-Conditioned Candidate Scoring

From Eq. (2), the topology descriptor is built as

$$Z_{\text{bus}} = Y_{\text{bus}}^+, \quad d_{ab}^Z = |Z_{aa} + Z_{bb} - Z_{ab} - Z_{ba}|. \quad (23)$$

For a BUS candidate c ,

$$\psi_Z(c) = [d_{cp}^Z]_{p \in \mathcal{P}}. \quad (24)$$

For a LINE candidate $\ell = (i, j)$,

$$d^Z(\ell, p) = \min\{d_{ip}^Z, d_{jp}^Z\}, \quad \psi_Z(\ell) = [d^Z(\ell, p)]_{p \in \mathcal{P}}. \quad (25)$$

A fixed training-side normalization operator $\mathcal{S}_Z$ gives $\tilde{d}^Z = \mathcal{S}_Z(d^Z)$, and the PMU-wise interaction block is

$$\mathbf{x}_{c,t}^Z = \left[\mathbf{x}_{p,t} \exp(-\tilde{d}^Z(c, p)) \right]_{p \in \mathcal{P}}. \quad (26)$$

For each target-active training row with true source c_t^* , the candidate label is

$$y_{c,t} = \mathbf{1}\{c = c_t^*\}. \quad (27)$$

Training uses the true candidate and a frozen subset of non-target candidates from the same admissible set; all admissible candidates are scored at inference.

$$s_\theta(c | \mathbf{x}_t) = h_\theta(\mathbf{x}_t, \text{type}(c), \psi_Z(c), \mathbf{x}_{c,t}^Z), \quad (28)$$

with

$$\hat{c}_t = \arg \max_{c \in \mathcal{C}_{\hat{c}_t}} s_\theta(c | \mathbf{x}_t). \quad (29)$$

The descriptor is physics guided rather than equation constrained. It embeds the network into the candidate representation but does not force the learner to satisfy the AC equations or the ANDES DAE during fitting. Because candidate-row construction and representation change together with the impedance variables, the current comparison evaluates the complete candidate-ranking design, not d^Z in isolation.

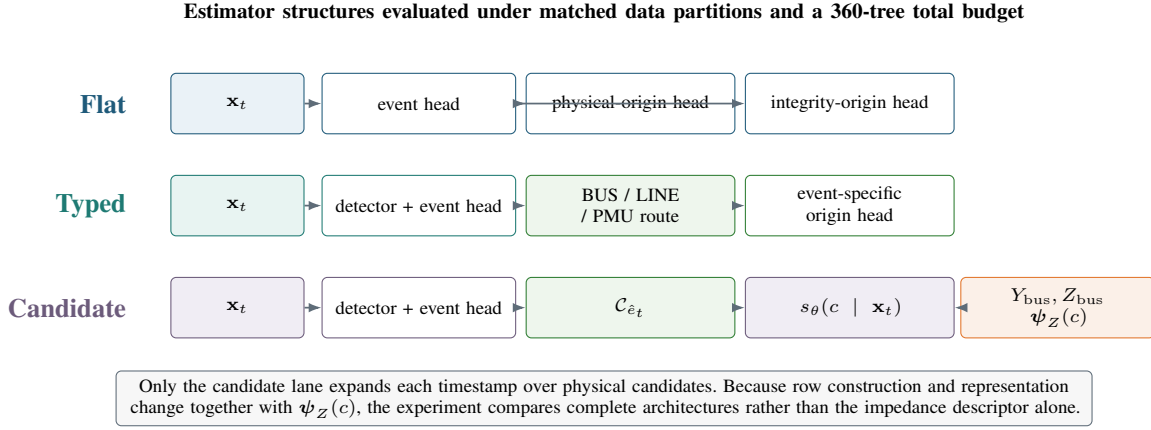


Fig. 4. Flat, typed, and candidate-conditioned estimator structures. Green denotes event-compatible routing and orange denotes network-derived candidate information; the other boxes are learned stages.

VI. EXPERIMENTAL DESIGN

A. Grouped Split and Statistical Controls

For every target key, three replicas enter training, one validation, and one test, yielding 414/138/138 whole-trajectory partitions. No trajectory contributes rows to more than one partition. The median imputer, tree ensembles, candidate-row construction, and training-side network-scaling constants are fitted or frozen from the training partition. Validation is used to select the abnormal decision threshold; the test trajectories are not accessed until final scoring. Because each target appears in every partition, the estimand is new-trajectory performance at known assets.

All variants use ExtraTrees with maximum depth 14, at least two samples per leaf, square-root feature subsampling, and a total budget of 360 trees. Equal total tree count controls one resource dimension but does not equalize per-head capacity because the architectures contain different numbers of heads.

The abnormal decision threshold is selected once from a 33-value validation grid spanning 0.1–0.9 and is then frozen for the test set. The resulting thresholds are 0.75 for flat and 0.45 for typed in all three model seeds.

B. Metrics and Uncertainty

Detection uses abnormal precision, recall, F1, row false-positive fraction, alarm episodes per normal-labeled minute, scenario detection, and aligned-sample detection delay. Event identification uses macro-F1 over labels 0–8. Localization uses row-level Top-1/Top-3 exact recovery. Scenario Top-1 is the modal physical prediction over target-active rows and is reported only as an offline aggregation; empty outputs count as errors.

Means and sample standard deviations summarize three model seeds. Paired intervals use 5000 bootstrap resamples of test scenarios within event family while preserving method-seed pairing. The intervals condition on this target-covered split; they do not represent uncertainty over held-out assets, PMU placements, or networks.

The candidate-support audit gives complete support for every evaluated typed candidate set because all target keys are present in training. Electrical-distance localization error is not available

in the current result set, so the architecture comparison uses Top-1 and Top-3 physical-location accuracy.

C. Timing

Model-only timing was measured on Windows 11 with an Intel Core Ultra 9 185H, 16 physical and 22 logical cores, and 31.5 GB RAM using `n_jobs=-1`. Each seed was timed once without warm-up. The measurement excludes ingestion, alignment, feature computation, alarm aggregation, and output, and is interpreted only as descriptive batch-scoring time.

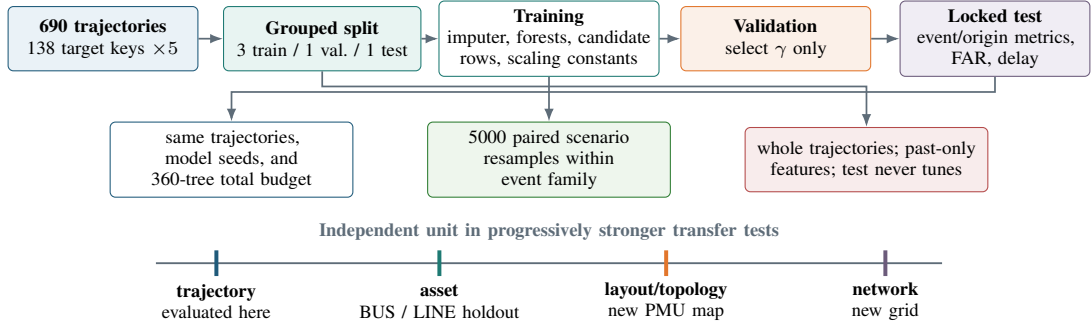


Fig. 5. Data partition, fitting boundary, and uncertainty protocol. All data-dependent preprocessing and model fitting are confined to training; validation selects the abnormal threshold; the test set remains untouched until final scoring.

VII. RESULTS

A. Primary Accuracy Results

Flat obtains 0.995 ± 0.0004 abnormal precision, 0.959 ± 0.001 recall, and 0.977 ± 0.001 F1. Typed obtains 0.981 ± 0.001 , 0.951 ± 0.002 , and 0.965 ± 0.002 , respectively. The paired interval for typed minus flat detection F1 is $[-0.0207, -0.0045]$.

Row-level physical Top-1 is 0.535 ± 0.012 for flat and 0.496 ± 0.017 for typed. The paired interval is $[-0.0772, -0.0029]$. Offline scenario aggregation gives 0.614 ± 0.029 for typed and 0.590 ± 0.017 for flat; the interval $[-0.0551, 0.1019]$ includes zero. The candidate ranker obtains 0.462 physical Top-1 by row and 0.581 by scenario. Relative to typed, the paired intervals are $[-0.0734, 0.0047]$ and $[-0.1129, 0.0441]$.

B. Resource Cost and Detection Timing

The serialized estimator bundle occupies 98.3 MB for the flat architecture and 28.1 MB for the typed architecture, a 71% reduction; model-only scoring time decreases from 0.051 to 0.025 s per input minute. The same 360-tree total is distributed across a different number of heads, so this is an accuracy-resource comparison rather than an equal-capacity comparison. Candidate expansion raises model-only scoring time to 0.586 s per input minute.

Flat detects 130 of 137 abnormal test scenarios with a mean aligned-sample delay of 0.0189 ± 0.0008 s; typed detects 129 with 0.0287 ± 0.0035 s. These values exclude communication, PMU/PDC, feature-computation, and alarm-aggregation latency.

C. Event-Family Diagnostics

Typed labels every missing-only test row as normal, while flat recall for that class is 0.006 ± 0.006 . Load localization remains below 0.05 for all three methods. With 18 admissible load buses, uniform Top-1 selection gives approximately $1/18 = 0.056$, placing the load result near chance.

Relative to typed class heads, the candidate-ranking design changes generation Top-1 from 0.188 to 0.328, missing-plus-physical from 0.741 to 0.804, mixed-event physical Top-1 from 0.822 to 0.860, and line-outage Top-1 from 0.658 to 0.507. Line outages account for 3128 of the 7915 physical test rows.

TABLE VIII

TARGET-COVERED TEST RESULTS ACROSS THREE MODEL SEEDS. S@1 IS OFFLINE MODAL PHYSICAL TOP-1 PER SCENARIO; FAR IS ALARM EPISODES PER NORMAL-LABELED MINUTE. TIMING EXCLUDES FEATURE COMPUTATION.

Method	Det. P	Det. R	Det. F1	Event M-F1	Phys. S@1	FAR	Phys. @1/@3	Integ. @1/@3	s/min	Nodes (k)	Size (MB)
Flat	.995 ± .000	.959 ± .001	.977 ± .001	.800 ± .006	.590 ± .017	3.22 ± .14	.535/.671	.888/.888	.051	282	98.3
Typed	.981 ± .001	.951 ± .002	.965 ± .002	.791 ± .011	.614 ± .029	4.01 ± .40	.496/.646	.870/.874	.025	176	28.1
Candidate ranker	.981 ± .001	.951 ± .002	.965 ± .002	.791 ± .011	.581 ± .029	4.01 ± .40	.462/.622	.874/.876	.586	290	25.7

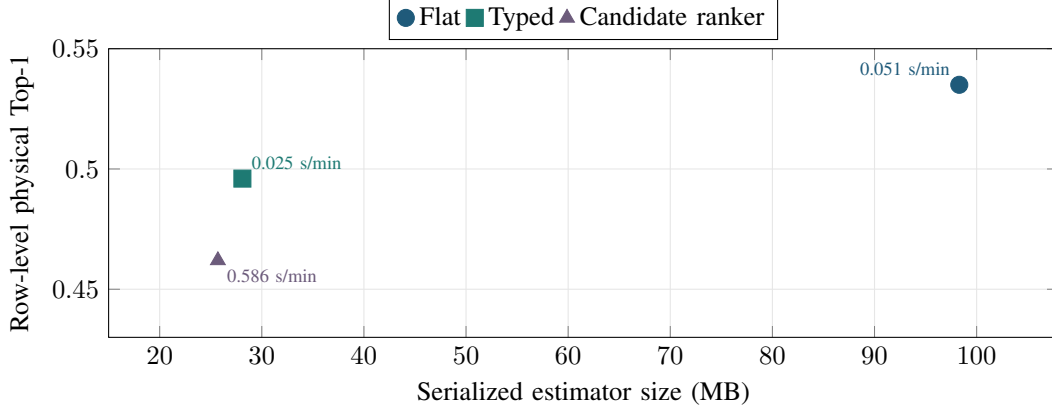


Fig. 6. Accuracy–resource trade-off for the three evaluated architectures. Typed routing reduces serialized size and model-only scoring time relative to flat heads, but the row-level localization point estimate is lower. Candidate expansion is compact on disk but slower at inference because each timestamp is expanded over candidate rows.

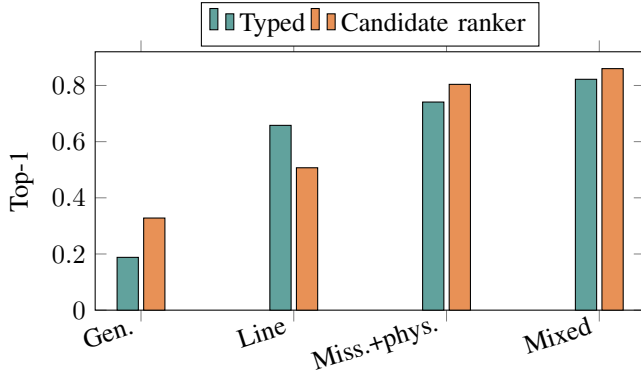


Fig. 7. Event-family physical Top-1 for typed class heads and the candidate-ranking design. The point estimates increase for generation, missing-plus-physical, and mixed events and decrease for line outages; the comparison does not isolate the impedance descriptor from the changed candidate representation.

VIII. DISCUSSION

A. Typed Routing Trades Accuracy for Compactness

Typed routing does not improve row-level localization in this experiment. Flat achieves 0.535 physical Top-1 versus 0.496 for typed, and flat also has higher detection F1. The measurable benefit is structural and computational: incompatible BUS, LINE, and PMU origins are never compared inside one location head, serialized size falls from 98.3 to 28.1 MB, and model-only scoring time falls from 0.051 to 0.025 s per input minute. Typed routing is therefore an accuracy–resource and output-semantics trade-off, not an accuracy improvement.

B. Static Impedance Conditioning Is Event Dependent

Candidate ranking gives 0.462 row-level physical Top-1 and 0.581 offline scenario Top-1, with paired intervals that do

not show an advantage over typed class heads. The family-level results are more informative: generation, missing-plus-physical, and the mixed class have higher point estimates, whereas line-outage Top-1 falls from 0.658 to 0.507. This pattern is consistent with the physical mismatch already visible in Section III: a static bus-impedance proximity is closer to an injection-to-voltage relation than to a line-outage redistribution signature. Because candidate row construction changes together with the network descriptor, the experiment cannot attribute these differences to d^Z alone.

C. Integrity and Load Cases Expose the Remaining Limits

The typed model labels all missing-only rows as normal even though missingness is explicitly represented. A deterministic availability signal is therefore being underused by the learned abnormal score. Load localization is also below 0.05 Top-1, close to the $1/18 = 0.056$ uniform baseline. The current results do not distinguish whether this is caused by weak features, insufficient operating-point variation, or poor source separability under the eight-PMU placement.

The alarm rate is also too high for an operator-facing interpretation: flat and typed produce more than three alarm episodes per normal-labeled minute. Any deployment study would need persistence or hysteresis logic and must report the resulting FAR–recall–delay trade-off. The timing numbers reported here cover model scoring only and exclude feature computation, communication, PMU/PDC processing, and alarm aggregation.

IX. LIMITATIONS AND FUTURE WORK

Every source asset appears during fitting, all event onsets are fixed at 2 s, and the study uses one network and one PMU

placement. The simulator output also omits the full PMU/PDC communication chain. Three model seeds on one grouped split quantify model variation but not target-, placement-, or network-level uncertainty.

The first unresolved question is the network descriptor itself. A fixed candidate scorer can compare absent, correct, and deliberately permuted descriptors with candidate rows, negatives, tuning, and compute held constant. Event-conditioned descriptors are the natural next comparison: transfer-impedance features for faults, PTDF/LODF-type signatures for line outages, and operating-point-dependent injection sensitivities for generation and load events. Leave-BUS-out and leave-LINE-out folds are then needed to test whether a shared candidate scorer can localize a source that never appears during fitting.

The measurement model also needs a controlled extension. Current phasors and ROCOF are available in the PMU contract and were used by the competition prototype but are absent from the 136-variable formulation evaluated here. Their incremental value should be measured with the same non-anticipative protocol. Randomized onset, broader operating points, multiple PMU layouts, longer normal streams, and a second transmission network would address external validity. Events 5 and 6 also motivate a deterministic availability gate, while future localization reports should include per-event confusion, electrical-distance error, and oracle-event routing alongside Top-1/Top-3.

X. CONCLUSION

Under a strict non-anticipative protocol, the three estimator structures lead to different trade-offs rather than a single accuracy ranking. Flat heads retain the best detection F1 and row-level physical Top-1. Typed routing cuts serialized size and model-only scoring time substantially, but with lower row-level accuracy. Candidate-conditioned localization does not improve the aggregate metric and is particularly weak for line outages, where static impedance proximity is a poor surrogate for redistribution physics. Missing-only detection and load localization remain unresolved.

The experiment supports a narrower conclusion: sparse source localization is controlled jointly by event mechanism, sensor placement, and evaluation unit. Stronger evidence for network-conditioned localization requires an isolated descriptor ablation and source-asset holdouts; operational evaluation additionally requires a realistic alarm budget and a fuller measurement chain.

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